Computer Graphics Final Project 2026

Theme: AI Future World

Design a 3D scene that represents an AI-powered future world. Your scene should include various animated objects/shapes, textures, lighting effects, and at least one AI-related character or object. (You may use the given SphereWorld project as a reference, but your scene should not look similar.)

Grading:

a. Scene Creativity 20%
b. Technique Difficulty 20%
c. Required Features (Item 2–6) 50%
d. Code, Report & Demo Video 10%

Project Requirement

1. **Texture Mapping (10%):**
   Use **4 or more textures**.

- Textures cannot be the same as course examples.
- Apply textures to different types of objects.
- At least one texture must be used on an AI-related object.

2. **AI theme-related Object Model (10%)**
   Load at least **one external OBJ model** related to AI or future technology.

- Cannot use provided example models.
- Must include animation.

3. **Animation (20%)**

   a. Self Rotation (5%)
   Object_A rotates on its own axis.
   b. Orbit Motion (5%)
   Object_B revolves around Object_A with a cosine/sine wave movement.
   c. Custom Animation (10%)
   Create at least two additional animation.

4. **Light and Shadow (10%):** Implement lighting and shadow effects.

- At least 2 light sources.
- Shadow must be visible.

Submission (10%)

Submit:

- Source code
- Texture images
- OBJ models
- PDF report
- Demo video

[PDF Report Requirements]

i. Project Setup

Explain how to compile and run your project.

ii. User Guide

Explain all keyboard and menu controls.

iii. Screenshots

Provide screenshots of your scene.

iv. Requirement Checklist

For each required item, clearly indicate:

- Which object fulfills the requirement.
- Screenshot evidence.

Failure to clearly identify completed requirements may result in point deductions.

v. Technical Challenges

Describe:

- Difficulties encountered.
- Solutions adopted.

[Demo Video requirement]

Provide a demonstration video showing:

- Scene overview
- Animation
- Lighting
- Shadow
- User interaction

Note

Do not use external game engines (Unity, Unreal Engine, etc.).

Only use OpenGL and the libraries provided in class, plus image processing libraries such as ~~OpenCV~~ if needed.

Submission deadline: 2026/6/29 11:59pm